Activity 2:
Combining infinite sequences – part 1

CLAIM

Pat says that there are more negative and positive whole numbers than just positive whole numbers.

Is Pat right?

Task A: Negating each term in a sequence

Train a robot called `Negator` that changes the sign of the number. For example, when `Negator`'s box looks like

it should give -3 to the bird. Try your robot out with different input numbers.

If you have troubles training the `Negator` robot then look at the pictorial instructions at the end of this activity sheet.

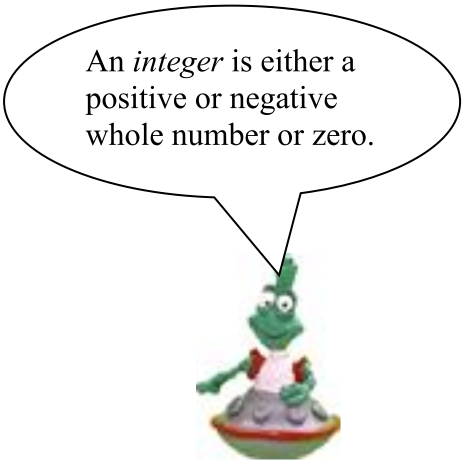
Connect `Negator` to the sequence of natural numbers made by the `Add 1` robot from Activity 1. Robots are connected by birds and nests.

Will there ever be a positive number on the nest that receives numbers from the `Negator` robot?

Explain…

Are there any negative *integers* that won't be put on the nest if the robots run forever?

Explain…



Task B: Turning two sequences into one

Train a robot called `Merge` *that turns two sequences into one sequence. For example, when* `Merge` *has a box like this*

`it` *should give 1 and then 2 to the bird. If you need help see the pictorial instructions at the end.*

Give `Merge` *the box*

where the first nest will receive all the positive integers and the second nest will receive all the negative integers.

Will every integer come out of `Merge` were it to run forever?

If any integers are missing on the output nest what could we do?

Task C: Integers dancing with *natural numbers*

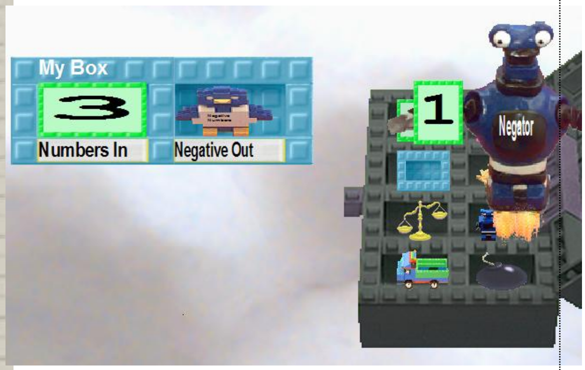
When you ran `Merge` on the positive and the negative *integers* you produced all the *integers* (except 0).

Can you see a way to get all the *natural numbers* to dance with all the integers?

Explain.

What about 0: who is it dancing with?

Activity 2, Task A: How to Train the `Negator` Robot



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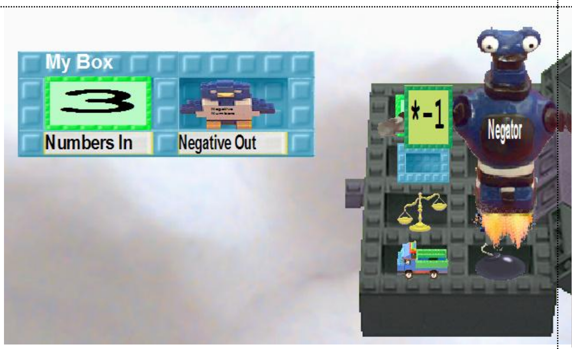

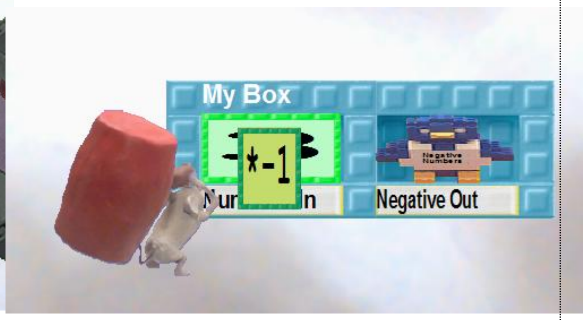

Activity 2, Task B: How to Train the `Merge` Robot